Field Studies and Data Analysis, Interpretation, and Presentation: Types of Analysis, Grounded Theory, and Frameworks

Week 10 - Lecture 12

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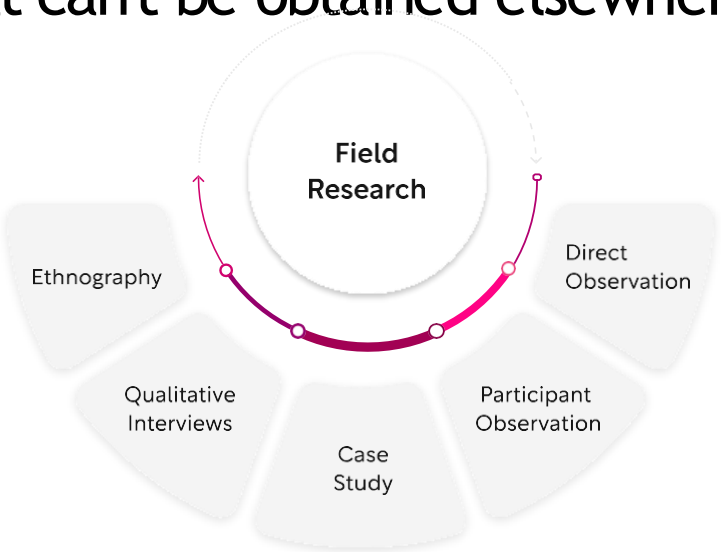
Introduction

- Advantages of observing users in their natural environment:
  - Provides more information about **real-life challenges**
  - Reflects a more **accurate** and **natural experience** using the product
  - Can observe **multi-tasking and distractions** that would not occur in a lab environment
- Field visits provide information about users' environment that can't be obtained elsewhere



Source: Wish Work

The Field Visit Process (1)

- Field observation can be enjoyable
- Field observation can also be exhausting
- Involves actively uncovering insights
- Involves writing and recording information
- Involves keeping the process moving
- Involves extracting interesting information from participants

The Field Visit Process (2)

Step 1 - Identify a Field Site

Step 2 - Establish connections with members on the field

Step 3 - Observe the subjects and their interaction with the surroundings

Step 4 - Collect data from the observations

Step 5 - Analyze data and generate working hypothesis

Step 6 - Focus on topic of interest and connect with Theory

Step 7 - Conduct interviews with focus groups

Step 8 - Complete the analysis and write a research study report

Source: Enago.com

The Field Visit Process (3)

- Remember the Small Details

  - Be **conscious** of being a visitor in **someone else's space**
  - Give participants an idea of **what to expect**
  - Be **friendly and empathetic** to put participants at ease
  - Ensure team members follow **site expectations**
  - Pay **attention to behaviors and expectations** on site (e.g. shoes off household)
  - **Write expectations** for product use
  - **Take note of unexpected situations**
  - Look for area of **further investigation**

Types of Field Observation (1)

- Several possible relationship models are listed in the textbook.

- Master/Apprentice:
- Observe the user as if you are learning from them.
  - Consider them the expert and focus on taking in everything they do.
  - Ask questions or encourage them to narrate their actions with a focus on understanding what is happening.

Types of Field Observation (2)

- Partnership:
  - Approach the activity or process as a joint discovery.
  - Work together to discover more about the ways of working with the product or process/service.
  - Ask questions that bring problems to the surface as they arise
  - Give the research participant space to talk about what they are doing and why.

Types of Field Observation (3)

- Interviewer/Interviewee:
  - Avoid asking questions that only reveal information.
  - Encourage the participant to talk through their experience.
  - Don't jump in with questions right away.
  - Ask questions to know what they perceive as significant or notable.
  - Let them offer the information on their own.

Types of Field Observation (4)

- Expert/Novice:
  - Consider yourself a novice.
  - Let the research participant know that you are there to understand their experience or practice.
  - Don’t intervene to solve their problems.
  - Observe the problems to inform your research.

How to Observe - More general guidelines

- Learning from the environment
  - Document your observations
  - Consider light levels, temperature, dust, dirt, moisture
  - People's relationships with each other and the objects around them

Field Research in Action: Summary

- Video

https://www.youtube.com/watch?v=bjZjO9D26Jk&ab_channel=KevinLiang

Remote Field Studies

- Video

https://www.youtube.com/watch?v=smzwIsi2Zto&tab_channel=NNgroup

Understanding Differences

- Video: Field Studies vs. Ethnographic Studies vs. Contextual Inquiry

https://www.youtube.com/watch?v=ULrHIFkwDqM&ab_channel=NNgroup

Analysis: Which Analytical Framework?

Framework	Data	Focus	Expected outcomes	Level of granularity
Conversation analysis	Recordings of spoken conversations	How conversations are conducted	Insights into how conversations are managed and how they progress	Word-level, or finer, for instance, pauses and inflection
Discourse analysis	Recordings of speech or writing from individuals or several participants	How words are used to convey meaning	Implicit or hidden meanings in texts	Word, phrase, or sentence-level
Content analysis	Any form of "text" including written pieces, video and audio recordings, or photographs	How often something is featured or is spoken about	Frequency of items appearing in a text	A wide range of levels from words, to feelings or attitudes, to artifacts or people
Interaction analysis	Video recordings of a naturally-occurring activity	Verbal and non-verbal interactions between people and artifacts	Insights about how knowledge and action are used within an activity	At the level of artifact, dialogue, and gesture
Grounded theory	Empirical data of any kind	Constructing a theory around the phenomenon of interest	A theory grounded in empirical data	Varying levels, depending on the phenomenon of interest
Systems-based frameworks	Large-scale and heterogeneous data	Large-scale involving people and technology, such as a hospital or airport	Insights about organizational effectiveness and efficiency	Macro-level, organizational level

Conversation Analysis - Martin Porcheron et al. (2018)

- Examines the semantics of a conversation in fine detail

- An extract of the conversation between a family and Alexa

```
01  SUS  i'd like to play beat the intro in a minute
02  LIA  [ oh no:: ]
03  SUS  [ alexa   ][ (1.1)  ] beat the in[tro
04  CAR             [ °yeah° ]
05  LIA                                   [°no:::...°
06  CAR  (0.6) it's mother's day? (0.4)
07  SUS  it's (    ) yep (.) listen (.) you need to keep
08       on eating your orange stuff (.) liam
09       (0.7)
10  CAR  and your green stuff
11  SUS  alexa (1.3) alexa (0.5)=
12  CAR                         =°and your brown stuff°
13  SUS  play beat the intro
```

Discourse Analysis

- Focuses on dialogue; that is, the meaning of what is said and how words convey meaning
- Language is viewed as a constructive tool
- Discourse analysis is useful when trying to identify subtle meaning

Disclosure Analysis - Example

Discourse analysis is what you do when you are saying that you are doing discourse analysis...

According to Coyle, discourse analysis is what you do when you are saying that you are doing discourse analysis...

Discourse Analysis

- Video: https://www.youtube.com/watch?v=R2o743ikYeY&ab_channel=EvanAshworth

Content Analysis

- Involves classifying data into themes or categories and studying their frequencies
- Can be used for any "text": video, newspapers, advertisements, images, and sounds
- Often used in conjunction with other techniques

Content Analysis (Examples)

For example, Yolande Strengers et al. (2020) focused on how visions for the smart home were being discussed in consumer, news, and trade articles. They analyzed 270 articles, including 166 feature images, published between 2000 and 2016 to understand the lifestyle expectations they promoted. Their findings indicate that the visions promote an enhanced quality of life including effortless energy-saving but also new opportunities to consume energy.

Julia Himmelsbach et al.'s (2019) study of diversity in HCI studies focused on frequency counts. Using predefined frameworks of diversity dimensions, they examined articles from CHI 2006, 2011, and 2016 to see how consideration of diversity in HCI studies has changed over this period. Their results showed that research still examined a limited number and set of dimensions, for example, the dimensions of Class and Education changed from 0.38 to 0.51 mean occurrences, and 1.46 to 1.33 mean occurrences, respectively, while sexual orientation increased from 0 to 0.07 mean occurrences.

Interaction Analysis

- A way to investigate and understand interactions between people, artifacts, and technologies
- Based on empirical observations such as videos
- Inductive process in teams, collaboratively
- Contents of the material is logged
- Materials are extracted, classified, or removed
- Instances of a salient event are assembled and played one after the other

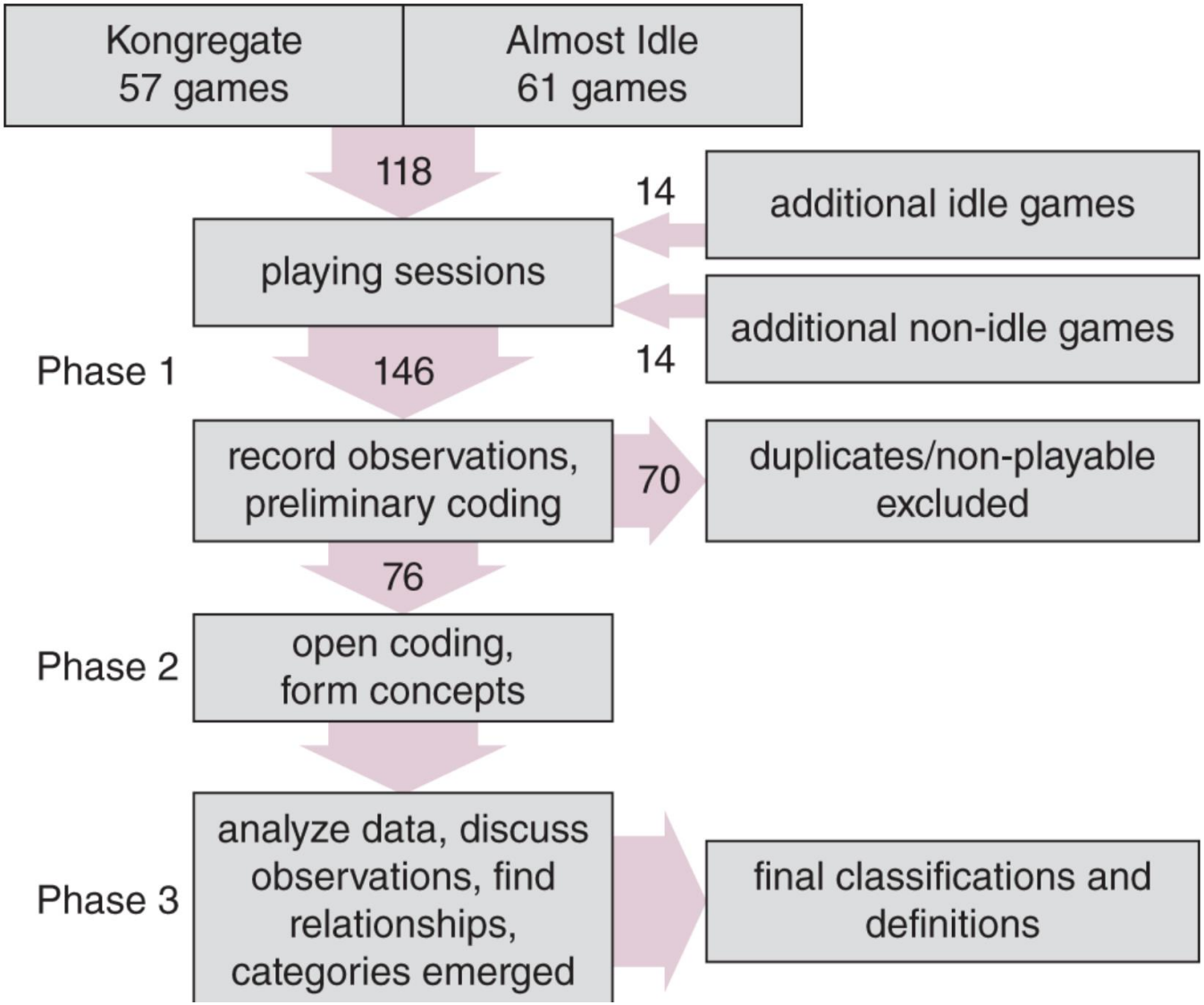
Interaction Analysis: Example

An example of interaction analysis being used in HCI is Naomi Polinsky et al.'s (2021) study of how parents and children collaborate when programming a robot to visit a series of locations on a map. They were interested in how parents and children supported each other through this form of communication. Specifically, they were interested in the symbols used and what elements of those symbols were most salient for their collaborative work. The pairs' coding interactions were recorded through two channels: a tablet that was used to program the robot, and a separate video recording of the pairs' activities. For analysis, these recordings were synced. Following the steps of interaction analysis, Naomi Polinsky and her colleagues generated multi-modal transcripts of activity from the two recordings, selected two cases that illustrated challenges due to parents' and children's different interpretations of symbols, and analyzed the recordings to provide insights about these differences and how the challenges were overcome. One of their findings was that the physical representational tool supported collaboration better than the digital tool. From their analysis they developed guidelines for codable robot design: to provide representational tools that invite shared meaning making; and that the representational tools should support translation into robot instructions.

Grounded Theory

- Seeks to develop theory from systematic analysis of empirical data
- Three levels of 'coding'
- Open: Identify categories. Similar to TA process
- Axial: Flesh out and link to subcategories
- Selective: Form theoretical scheme

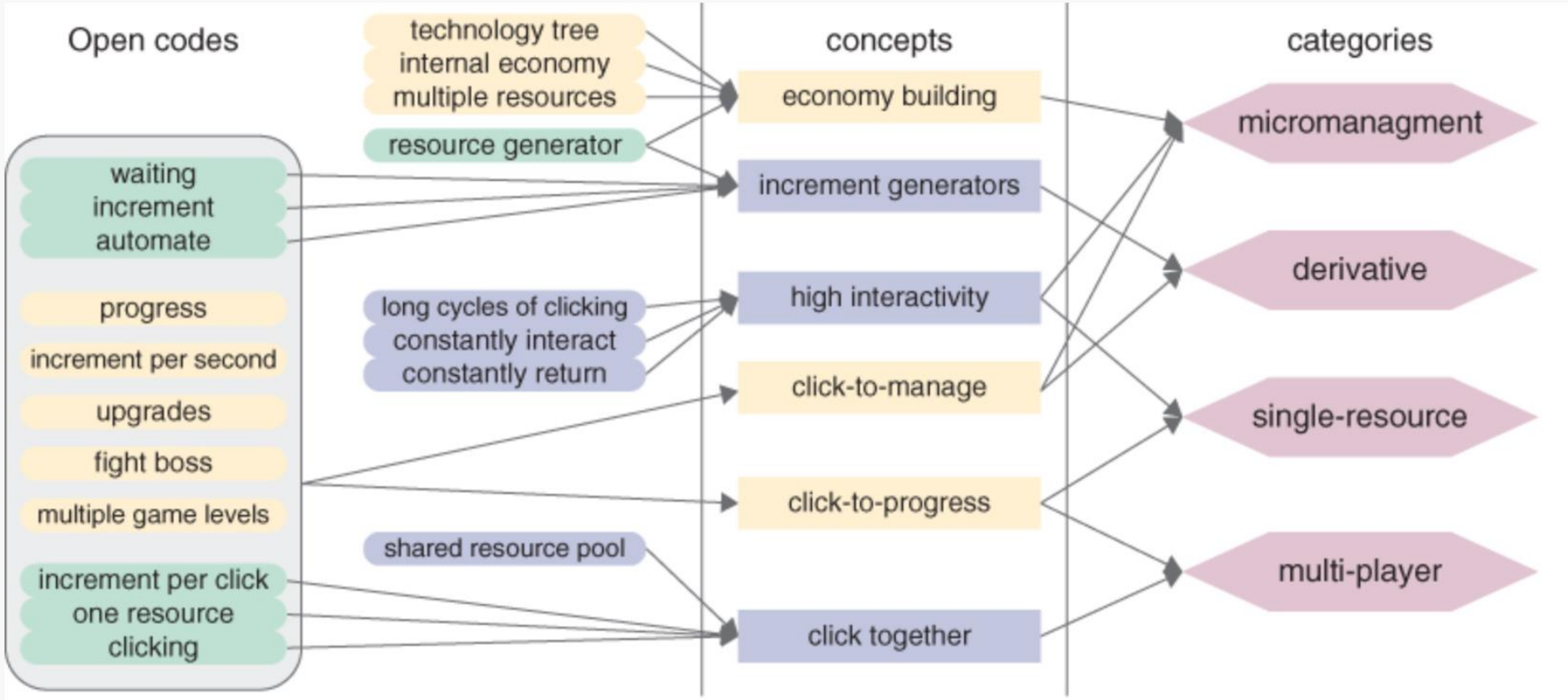
Grounded Theory – Example (Idle Games 1)



Grounded Theory – Example (Idle Games 2)

Game Feature	Observations
Game name	*AdVenture Capitalist* [G38]
Play description	You start CLICKING on a lemonade stand and collect money. Spend money to make upgrades, INCREASE PRODUCTION PER CLICK. Start hiring workers and INCREASE PRODUCTION PER SECOND. When you have enough money, you can buy new businesses, automate all your businesses to INCREMENT more money, and leave the game progress.
Game mechanics	Click to gain money, AUTOMATE production, make upgrades to DAMAGE/SEC.
Rewards	ONE CURRENCY, which is money, is rewarded in return.
Interface	GRAPHICAL
Interactivity level	7
Progress rate	9
Overview	This is a SINGLE-PLAYER game, which requires LONG CYCLES OF CLICKING at the start, and making a number of upgrades. Production rate reaches $390/sec in less than 10 minutes, and you gain 1M in cash making the game progress faster.

Grounded Theory – Example (Idle Games 3)

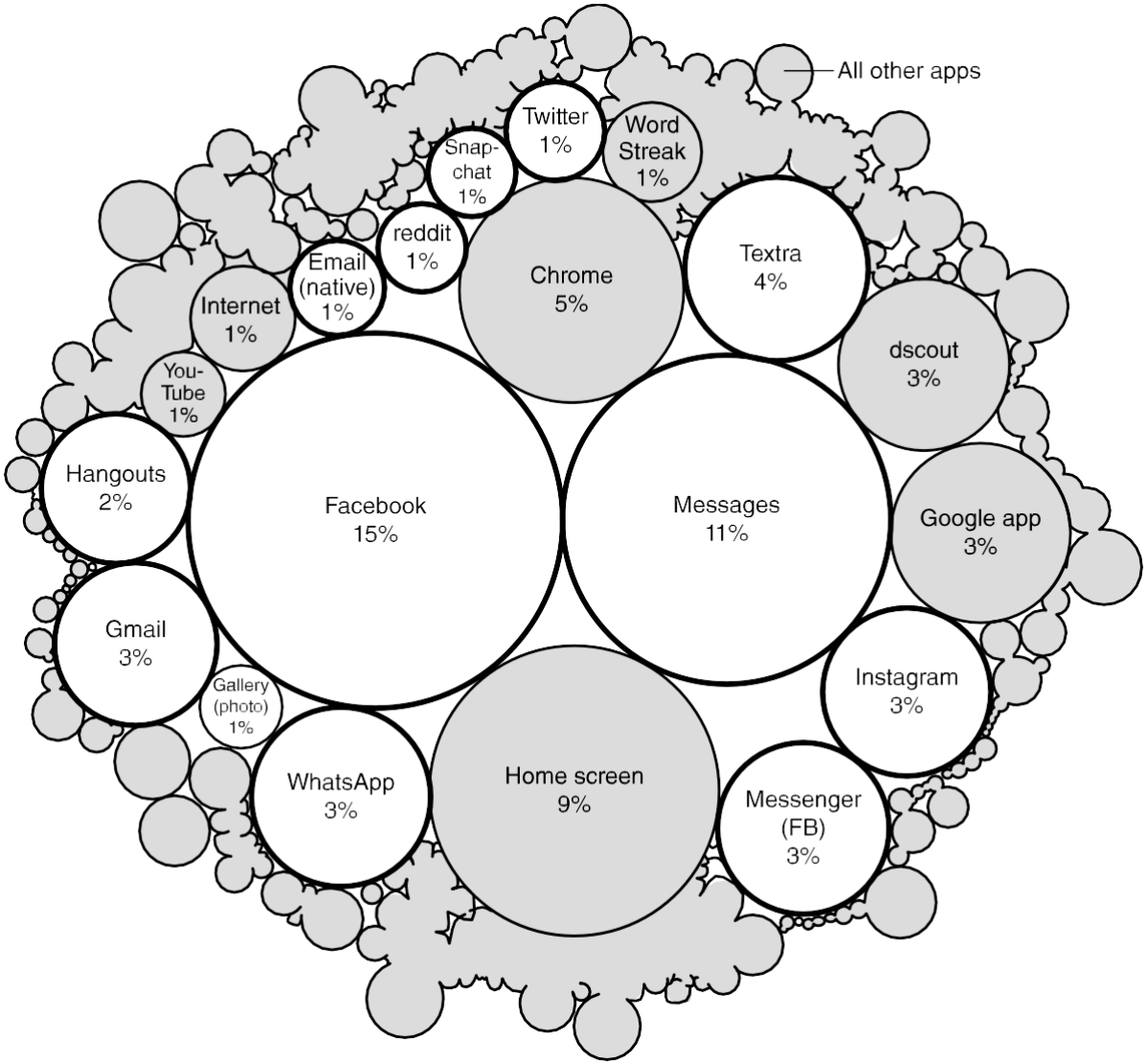
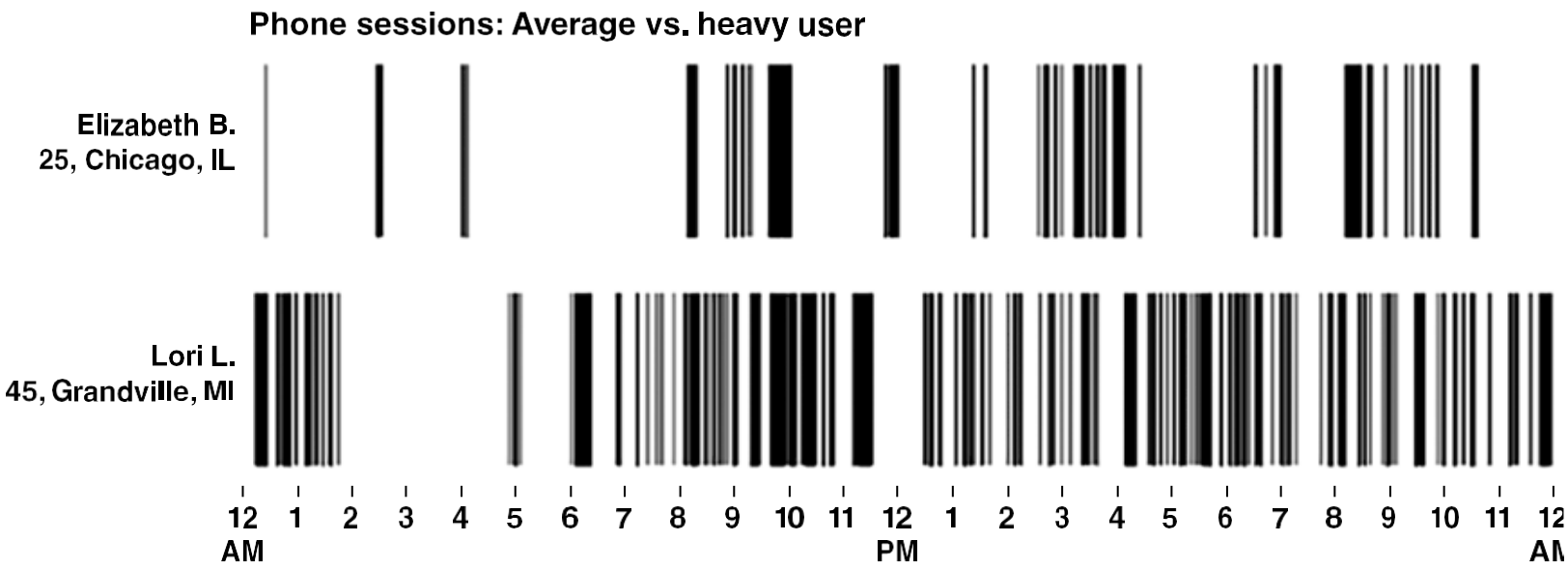


The analysis process that developed the incremental games super-category (each category above is part of incremental games). The process started with open coding of observations on idle games: multiple codes are created. Concepts are discovered through analyzing the open codes and identifying common features. This is an iterative process, where new codes are added, combined, or deleted. Each code is connected to one or more games and can be combined to form new concepts. Concepts are analyzed to find common relationships, and, thus, categories emerge. In the diagram, coloration is only to aid in reading. The left grouping is to show that all contained codes are part of click-to-manage and click-to-progress.

Tools to support data analysis

Two well-known tools that support some of these data analysis activities are **Nvivo** and **Dedoose**.
For example, Nvivo supports the annotation and coding of data including PDF documents, photos, and video and audio files. Using Nvivo, field notes can be searched for keywords or phrases to support coding or content analysis; codes and data can be explored, merged, and manipulated in several ways. The information can also be printed in a variety of forms such as a list of every occasion a word or phrase is used in the data, and a tree structure showing the relationships among codes. Like all software packages, Nvivo has advantages and disadvantages, but it is particularly powerful for handling large sets of data and can generate output for statistical packages.

Interpreting and presenting the findings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